The “multi-performance plot” in simulation studies:

a compact visualisation of up to seven performance measures comparing multiple statistical methods

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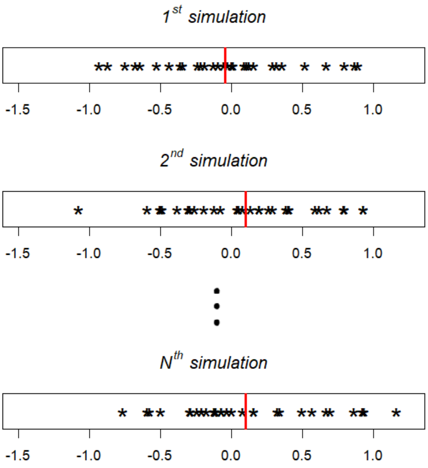
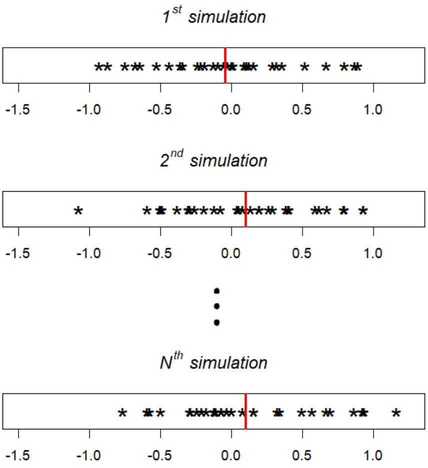
Setup



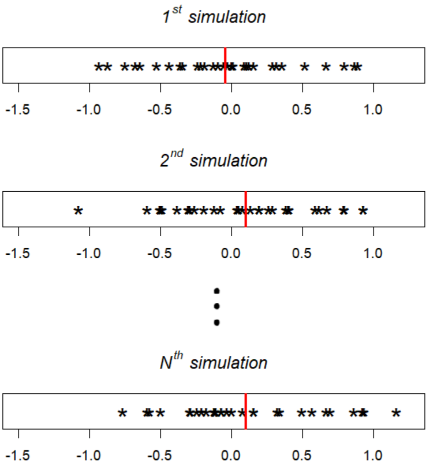
- θ = true value of parameter
- $\widehat{\theta}$ = estimator of θ
- $\widehat{\theta}_1, \widehat{\theta}_2, \cdots, \widehat{\theta}_N$ = estimates of θ
- N = number of simulations

Bias



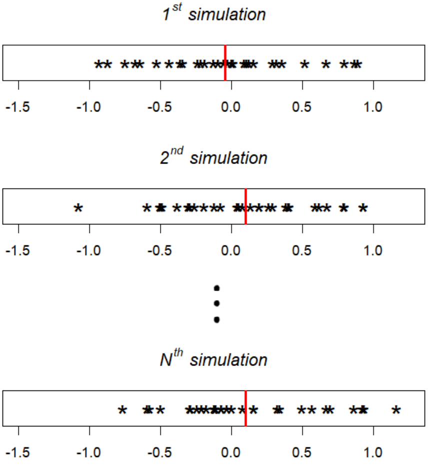
- $\text{Bias} = \mathbb{E}[\widehat{\theta}] - \theta$
- $\widehat{\text{Bias}} = \text{mean}(\widehat{\theta}_i) - \theta$

Empirical standard error



- EmpSE $= \sqrt{\mathbb{V}[\widehat{\theta}]}$
- $\widehat{\text{EmpSE}} = \text{sd}(\widehat{\theta}_i)$

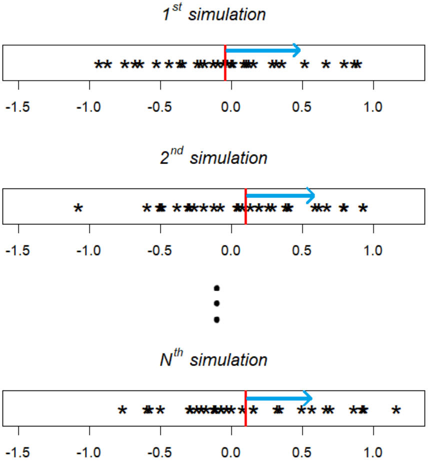
Mean squared error



- $\text{MSE} = \text{Bias}^2 + \text{EmpSE}^2$

- $\widehat{\text{MSE}} = \widehat{\text{Bias}}^2 + \left(1 - \frac{1}{N}\right) \widehat{\text{EmpSE}}^2$

Average model standard error



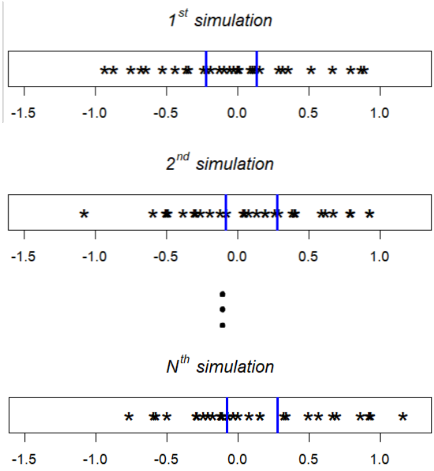
- $\text{ModSE} = \sqrt{\mathbb{E}[\widehat{\mathbb{V}}[\widehat{\theta}]]}$
- $\widehat{\text{ModSE}} = \sqrt{\text{mean}(v_i)}$ where

  v_i = estimated variance of $\widehat{\theta}_i$
- **Ideally $\widehat{\textbf{ModSE}} = \widehat{\textbf{EmpSE}}$**

  average of variances

  = variance of averages

Coverage



- Coverage $= \mathrm{P}((\widehat{\theta}^{-}, \widehat{\theta}^{+}) \ni \theta)$

- $\widehat{\text{Coverage}} = \dfrac{\#\{i : (\widehat{\theta}_i^{-}, \widehat{\theta}_i^{+}) \ni \theta\}}{N}$

Two other performance measures (Morris *et al.* [2019])

- *Relative error in ModSE* estimated by $\dfrac{\widehat{\text{ModSE}}}{\widehat{\text{EmpSE}}} - 1$

- *Relative increase in precision* estimated by $\dfrac{\widehat{\text{EmpSE}_A^2}}{\widehat{\text{EmpSE}_B^2}} - 1$

Forms of presentation

- Tables?

- Plots?

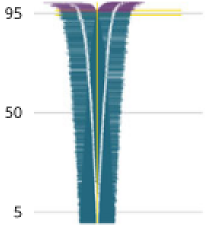
Forms of presentation: tables

θ	0					0.5					1.5				
Par.	**Bias**	**ESD**	**ASE**	**RMSE**	**CR**	**Bias**	**ESD**	**ASE**	**RMSE**	**CR**	**Bias**	**ESD**	**ASE**	**RMSE**	**CR**
	$\eta_3 = 1$														
β_0	0.000	0.130	0.130	0.130	0.948	−0.007	0.153	0.152	0.153	0.946	−0.019	0.165	0.165	0.166	0.938
β_1	−0.010	0.168	0.168	0.169	0.944	−0.017	0.177	0.177	0.178	0.940	−0.026	0.178	0.178	0.180	0.932
β_2	0.008	0.162	0.158	0.162	0.936	−0.000	0.180	0.174	0.179	0.936	−0.013	0.183	0.177	0.183	0.933
η_0	0.284	0.424	0.422	0.510	0.914	0.273	0.434	0.432	0.513	0.919	0.257	0.438	0.436	0.508	0.926
η_1	0.024	0.280	0.280	0.281	0.935	0.021	0.282	0.282	0.283	0.937	0.016	0.282	0.281	0.283	0.937
η_2	0.087	0.232	0.236	0.248	0.923	0.081	0.241	0.244	0.254	0.935	0.071	0.242	0.245	0.252	0.940
σ_T	0.008	0.096	0.094	0.096	0.946	0.002	0.109	0.106	0.109	0.945	−0.006	0.112	0.109	0.112	0.948
σ_C	0.002	0.204	0.200	0.204	0.933	−0.005	0.213	0.209	0.213	0.934	−0.016	0.215	0.210	0.215	0.932
ρ	−0.027	0.212	0.193	0.213	0.952	−0.028	0.215	0.194	0.217	0.951	−0.031	0.220	0.195	0.222	0.950
θ	−0.000	0.031	0.030	0.031	0.946	−0.005	0.062	0.060	0.062	0.948	−0.018	0.104	0.101	0.106	0.945
	$\eta_3 = 2$														
β_0	0.003	0.125	0.124	0.125	0.950	−0.004	0.150	0.149	0.150	0.947	−0.016	0.165	0.164	0.165	0.940
β_1	−0.010	0.160	0.161	0.160	0.947	−0.016	0.170	0.171	0.170	0.942	−0.026	0.172	0.173	0.174	0.933
β_2	0.010	0.159	0.153	0.159	0.941	0.001	0.179	0.173	0.178	0.938	−0.012	0.184	0.178	0.184	0.936
η_0	0.552	0.549	0.555	0.779	0.863	0.543	0.560	0.563	0.779	0.873	0.526	0.564	0.565	0.771	0.881
η_1	0.037	0.316	0.320	0.318	0.927	0.033	0.319	0.322	0.320	0.928	0.026	0.319	0.320	0.320	0.931
η_2	0.194	0.232	0.251	0.302	0.849	0.187	0.243	0.261	0.306	0.874	0.174	0.246	0.263	0.301	0.886
σ_T	0.007	0.093	0.093	0.093	0.947	0.001	0.107	0.107	0.107	0.950	−0.008	0.111	0.110	0.111	0.946
σ_C	0.052	0.250	0.249	0.255	0.923	0.045	0.259	0.256	0.262	0.925	0.034	0.261	0.258	0.263	0.928
ρ	−0.047	0.221	0.212	0.226	0.936	−0.050	0.226	0.214	0.231	0.933	−0.053	0.231	0.215	0.237	0.932
θ	0.000	0.030	0.030	0.030	0.949	−0.005	0.061	0.061	0.062	0.951	−0.017	0.105	0.103	0.107	0.950

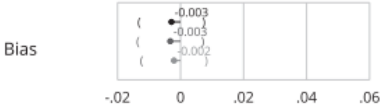
(Table 3 of Deresa and van Keilegom [2020])

Morris *et al.* [2019]

- *zip plot*: coverage only



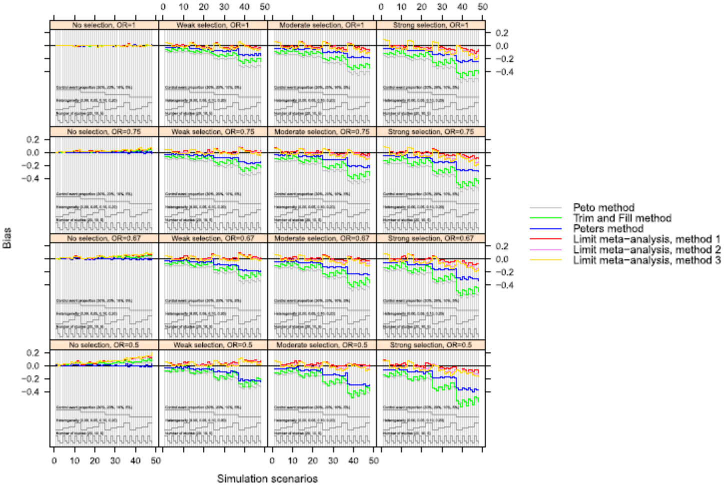
- *lollipop plot*: one performance measure per plot



Forms of presentation: single-performance plots

Rücker and Schwarzer [2014]

- *nested-loop plot*: many more combinations of simulation parameters



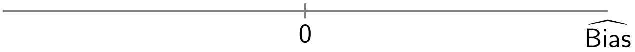
Objective

Simultaneous visualisation of

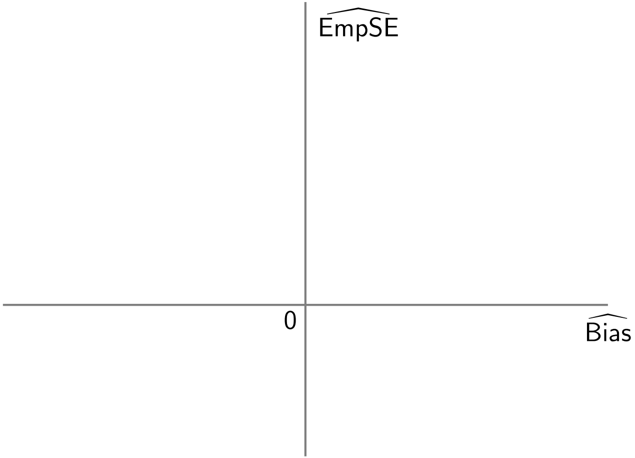
- all five performance measures (bias, EmpSE, MSE, ModSE, coverage);
- multiple combinations of simulation parameters;
- multiple statistical methods

in one plot

Multi-performance plot: design



Multi-performance plot: design



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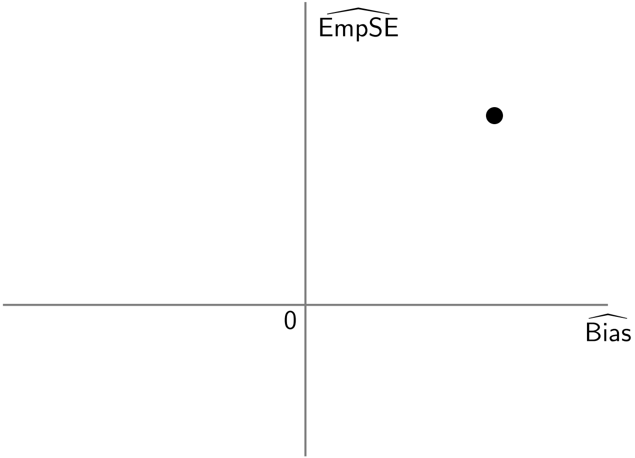

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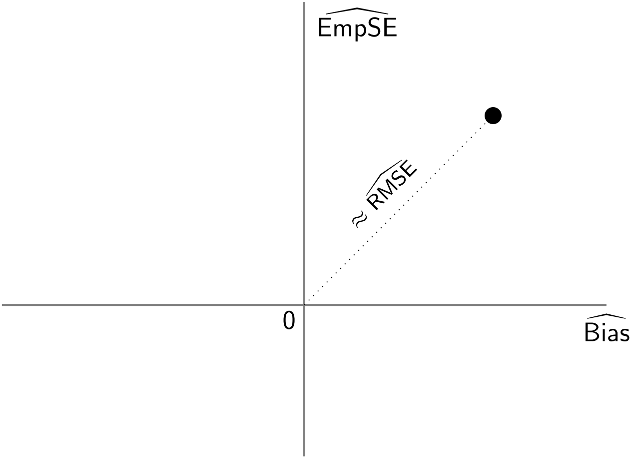
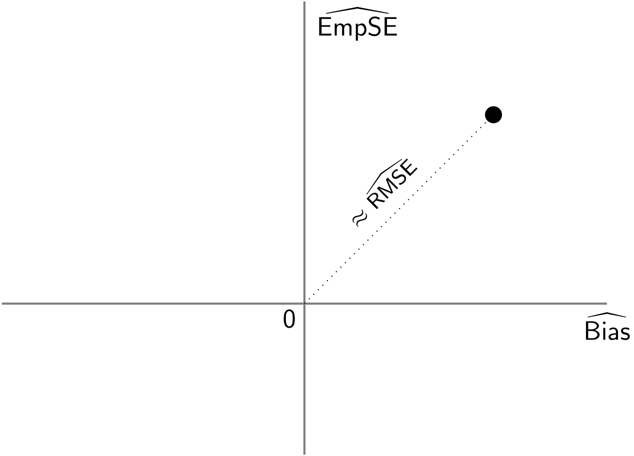
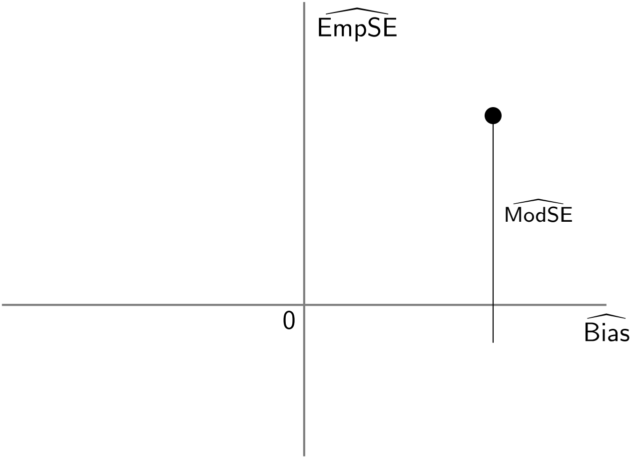


Multi-performance plot: design



Note: $\text{RMSE} = \sqrt{\text{Bias}^2 + \text{EmpSE}^2}$ but $\widehat{\text{RMSE}} = \sqrt{\widehat{\text{Bias}}^2 + \left(1 - \frac{1}{N}\right)\widehat{\text{EmpSE}}^2}$

Multi-performance plot: design



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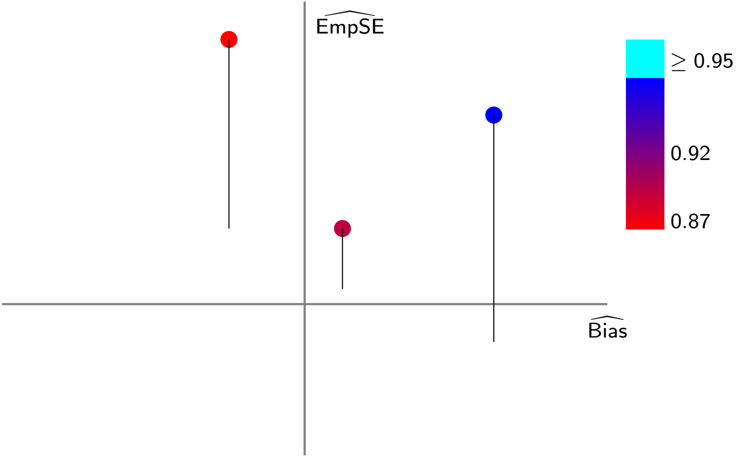

Coverage is visualised relatively:

- red-blue colour gradient if coverage < 0.95
- cyan if coverage ≥ 0.95

Multi-performance plot: Parast *et al.* [2016]

Background

- **PTE** $\in [0,1]$ measures how well a "substitute" endpoint can replace a patient-relevant endpoint in a clinical trial
- $\mathbf{R_S^M}$ and $\mathbf{R_S^F}$ are established methods to estimate the PTE
- Can their proposed method $\mathbf{R_S}$ outperform both methods, under the following full factorial design?

1. *statistical method* — R_S; R_S^M; R_S^F
2. *true value of PTE* — 0.2; 0.5; 0.9
3. *size of trial (per arm)* — 200; 1000
4. *model type* — correctly specified; misspecified

Multi-performance plot: Parast *et al.* [2016]

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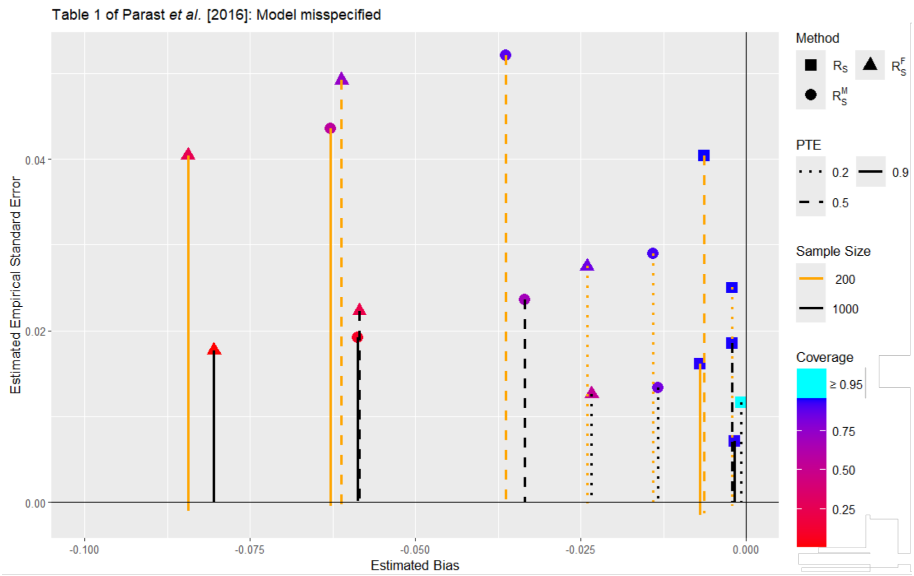
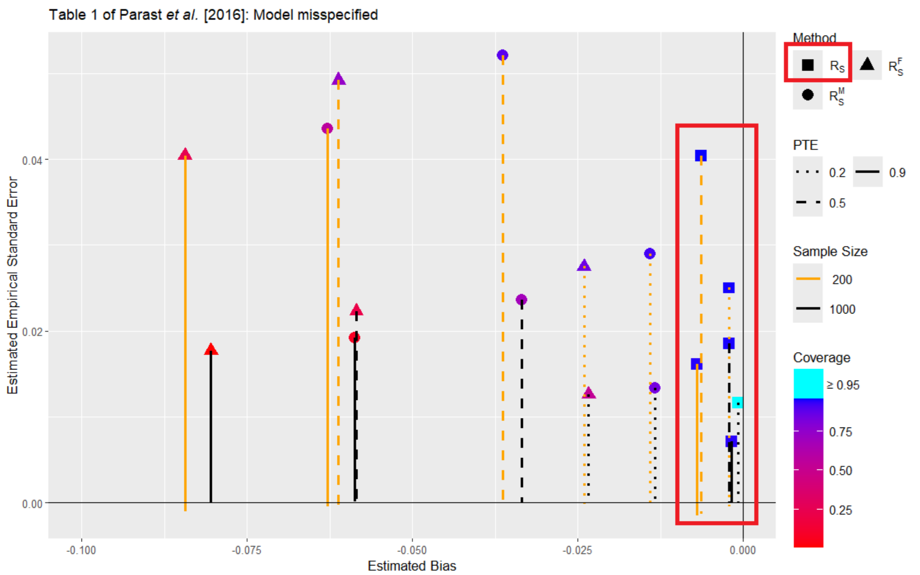
$(3 \times 3 \times 2 \times 1 = 18$ combinations$) \times 5$ performance measures $= 90$ values

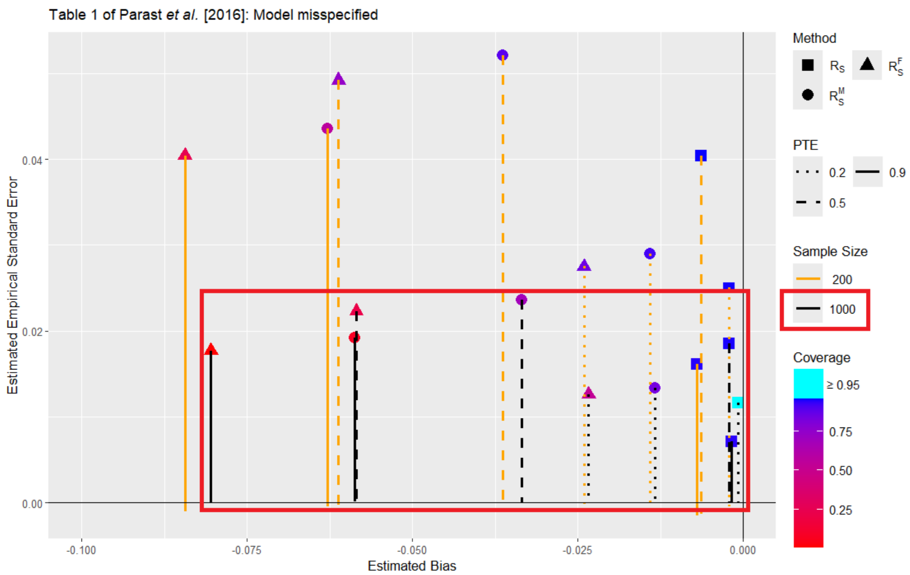
Multi-performance plot: Parast *et al.* [2016]



R_S outperforms R_S^M and R_S^F



Larger trial size preferable



Multi-performance plot: Deresa and van Keilegom [2020]

Background

- Survival time (T) is not always independent of censoring time (C)

- Data generation: $\begin{cases} \Lambda_\theta(T) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon_T \\ \Lambda_\theta(C) = \eta_0 + \eta_1 X_1 + \eta_2 X_2 + \varepsilon_C \end{cases}$

  where $\varepsilon_T \sim N(0, \sigma_T^2)$ and $\varepsilon_C \sim N(0, \sigma_C^2)$ with correlation ρ

- Estimation: existing "independent censoring model" assumes $\rho = 0$
  proposed "dependent censoring model" accounts for ρ

Multi-performance plot: Deresa and van Keilegom [2020]

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Multi-performance plot: Deresa and van Keilegom [2020]

How well does the dependent censoring model estimate

$$\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2); \quad \boldsymbol{\eta} = (\eta_0, \eta_1, \eta_2); \quad \boldsymbol{\sigma} = (\sigma_C, \sigma_T, \rho); \quad \theta$$

under the following full factorial design?

1	*statistical model*	dependent censoring; independent censoring
2	*censoring*	45%; 25%
3	*true value of* θ	0; 0.5; 1.5

Multi-performance plot: Deresa and van Keilegom [2020]

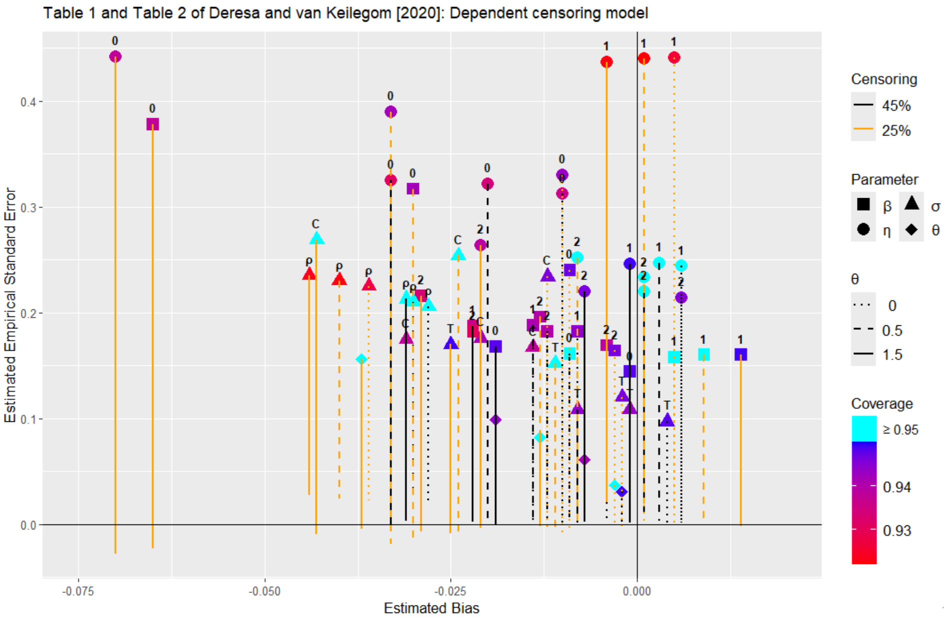
Model parameters β_0; β_1; β_2; η_0; η_1; η_2; σ_C; σ_T; σ_ρ; θ

1. *statistical model* dependent censoring; independent censoring
2. *censoring* 45%; 25%
3. *true value of θ* 0; 0.5; 1.5

10 estimands $\times$ $(1 \times 2 \times 3)$ combinations $\times$ 5 performance measures

$= 300$ values

Multi-performance plot: Deresa and van Keilegom [2020]



Summary

- Pattern identification from tabular presentations of performance measures can be difficult, especially in large simulation studies

- The multi-performance plot is proposed as a tool to assist with pattern identification
  - ✓ compactly displays dozens of performance measures
  - ✓ choice of shapes, colours and line types up to user
  - ✓ serves to complement tables, not to replace them
  - ✗ oversaturation can make identification harder!

Acknowledgements and Contact

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